



Prostate cancer biomarkers: from early diagnosis to precision treatment

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Abstract

Prostate cancer (PCa) is the second most prevalent cancer in men. In 2020, approximately 1,414,259 new cases were reported that accounted for 3,75,324 deaths (Sung et al. in CA 71:209–249, 2021). PCa is often asymptomatic at early stages; hence, routine screening and monitoring based on reliable biomarkers is crucial for early detection and assessment of cancer progression. Early diagnosis of disease is key step in reducing PCa-induced mortality. Biomarkers such as PSA have played vital role in reducing recent PCa deaths. Recent research has identified many other biomarkers and also refined PSA-based tests for non-invasive diagnosis of PCa in patients. Despite progress in screening methods, an important issue that influences treatment is heterogeneity of the cancer in different individuals, necessitating personalized treatment. Currently, focus is to identify biomarkers that can accurately diagnose PCa at early stage, indicate the stage of the disease, metastatic nature and chances of survival based on individual patient profile (Fig. 1).

Keywords Prostate cancer · Diagnosis · Biomarker · PSA · New generation biomarkers

Introduction

PCa incidence rates have gone up in the last decade, while PCa-induced mortality has decreased. This is mainly a result of PCa screening using PSA as a biomarker. Further risk stratification and refinement is achievable only using additional biomarkers. Sixty percent of cases are diagnosed in men aged 65 or older. Conservative treatment is typically recommended for the majority of PCa patients with slow disease progression. However, around 10% of patients develop metastatic disease, leading to a higher chance of PCa-related death [2]. PCa treatment is complex because of the heterogeneous nature of the disease, and accurate diagnosis of altered pathways leading to PCa is critical to ensure appropriate intervention. The identification of biomarkers for diagnosing PCa and predicting prognosis has greatly enhanced this area of medical research [2].

A biomarker is a measurable characteristic that provides information about normal and pathological processes, as well as the response to treatment [3]. It enhances clinical and pathological analysis by aiding in cancer screening, disease monitoring, prognosis prediction, and assessing treatment effectiveness. Biomarkers also play a crucial role in drug development and can potentially improve the safety and efficacy of cancer treatments by enabling personalized medicine approaches for patients [3].

Historically, a study from 1938 showed elevated prostatic acid phosphatase (PAP) in the prostates of patients with metastatic PCa. PAP was the first biomarker used routinely in diagnosing and staging of prostate cancer [4]. In the 1970s, prostate-specific antigen (PSA), also known as hK3, was discovered and today it remains the most widely used, though controversial, biomarker in prostate cancer [5, 6] (Fig. 2).

Currently for PCa screening, routine assessment procedures are followed (Fig. 3). These include the digital rectal examination (DRE), blood test to measure PSA levels, and transrectal ultrasound [7]. DRE is one of the fundamental tests for screening and early detection of PCa [8] (Fig. 3). Although it has an overall accuracy rate of approximately 59%, DRE remains a routine method due to its ability to detect cancers that may be missed by other diagnostic tests [9]. One significant advantage of DRE is its potential to

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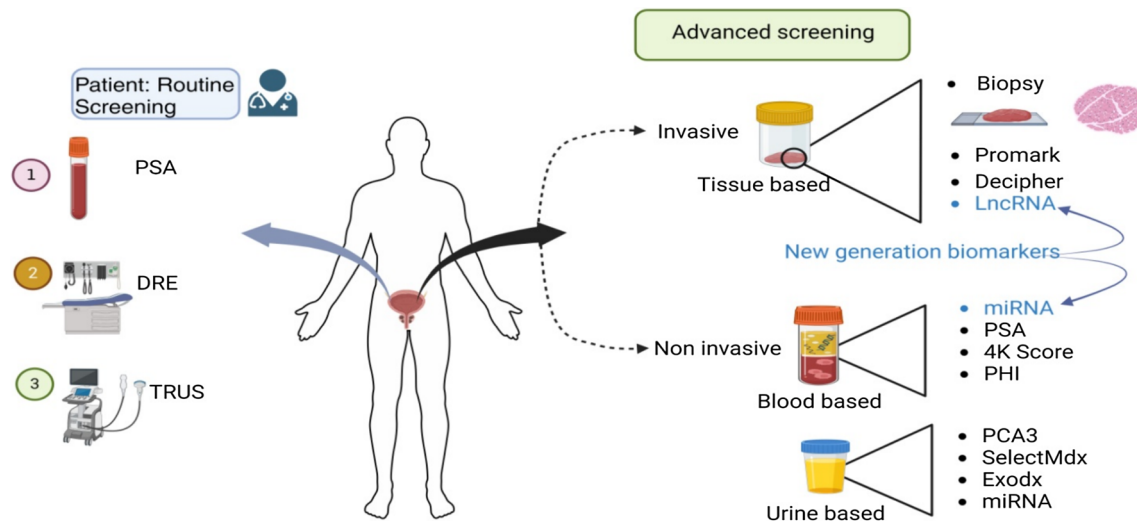


Fig. 1 Graphical abstract

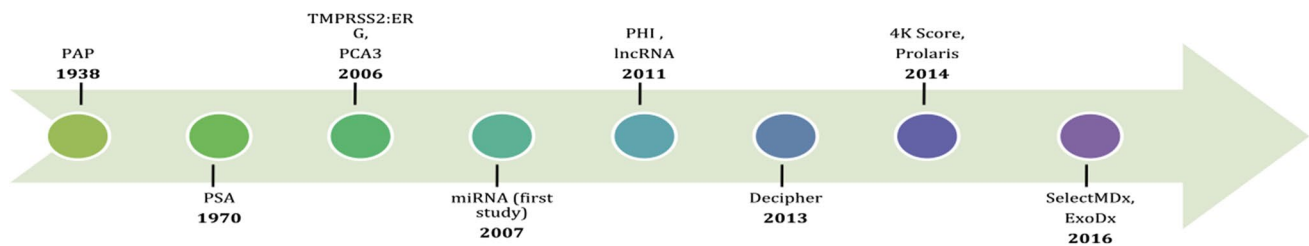


Fig. 2 Timeline showing PCa biomarker development

identify tumors in men with normal levels of PSA, particularly when the tumors are small and well differentiated [3]. However, DRE fails to detect early-stage PCa, as most noticeable tumors are already advanced. Further, many clinically significant tumors are located in areas of the prostate that are distant and therefore difficult to access through manual examination [9].

Discovered in 1971, PSA is widely recognized as a critical biomarker for the early detection, staging, and monitoring of PCa [10]. Similar to the digital rectal examination, PSA testing is a relatively affordable and widely accepted screening method. One of the primary advantages of PSA test is its high sensitivity. However, it must be noted that PSA testing lacks specificity, since other common conditions like benign prostatic hyperplasia and prostatitis can also yield abnormal PSA levels [11]. The detection of false-positive findings necessitates that individuals with high PSA undergo further diagnostic procedures, leading to increased costs and more invasive interventions [12] (Fig. 3). A more invasive, and more accurate diagnostic technique is provided by transrectal ultrasound-guided prostate biopsy. It is one of the primary methods to detect prostate cancer [13]. In this procedure, an ultrasound probe is inserted into the rectum

to examine and visualize the prostate, guiding biopsy needles through the rectum to collect tissue samples [13]. It is typically performed on an outpatient basis under local anesthesia, ensuring a safe and comfortable experience for patients [13].

To enhance specificity of the PSA test, researchers have proposed various modifications to the conventional test. For example the Prostate Health Index (PHI), 4 K score, and OncoType dx diagnostic techniques have been developed which take into consideration other genes and markers in addition to PSA (discussed in detail later) [14]. These tests aim to reduce the number of unnecessary prostate biopsies. Numerous studies have been conducted by scientists to correlate molecular biomarkers, such as androgen variant V7 [15] or inactivation of the PTEN or c-MYC gene [16] with disease in PCa patients, but none of them have been accepted as prognostic biomarkers for use in clinical trials [17].

Review of literature generated in the last decade shows that despite identification of different diagnostic and prognostic PCa biomarkers, PSA still continues to be the marker of choice for early detection and progression of PCa (Fig. 4). The reason for popularity of PSA test includes its organ-specific expression, ready availability in health care centers,

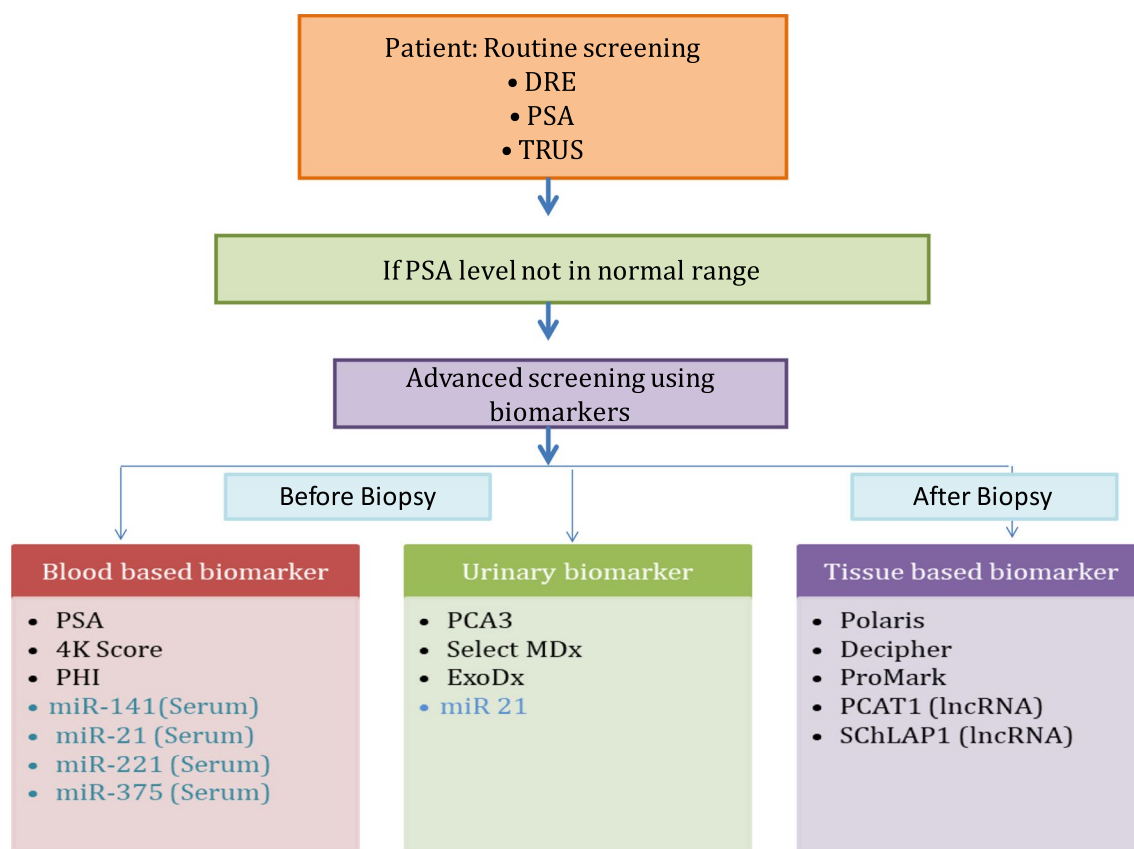


Fig. 3 Flow diagram showing screening of PCa

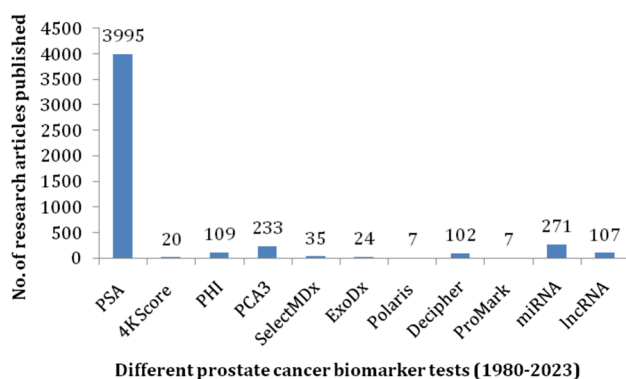


Fig. 4 A comparative analysis of the number of articles published in the last decade. NCBI database was searched as on 24 September 2023, to acquire the respective number of publications shown

and low cost- \$ 100 per test compared to \$ 3800–\$ 5000 per test for other diagnostic tests. Literature review also reveals that accurate diagnostic marker for early detection is still lacking significantly [18]. New generation markers such as miRNA, long non-coding RNA, and SNPs may provide increased sensitivity and specificity compared to PSA alone and help in early diagnosis, prognosis, reducing the number

of patient biopsy, and overall cost [19]. In addition, these biomarkers can assess molecular pathway dysregulations, enabling targeted therapies and personalized treatment strategies for patients.

Here in this review, we discuss different categories of PCa biomarkers along with a critical analysis of their advantages and disadvantages. In addition, we focus on new-generation biomarkers as tools for diagnostic accuracy and pathway-specific lineage for precision treatment.

Materials and methodology

Literature search was carried out about prostate cancer biomarkers in peer-reviewed journals between 2010 and 2023. Electronic databases including PubMed and Google Scholar were searched using appropriate keywords and Boolean operators. The following search terms were used: “prostate cancer biomarkers”, “PSA screening”, “diagnostic biomarkers”, “prognostic biomarkers”, “miRNA in prostate cancer”, “long non-coding RNA in prostate cancer”, “genetic markers in prostate cancer”, “prostate biopsy”, “transrectal ultrasound”, “Prostate Health Index”, “4 K score”, “OncoType dx”, and “personalized treatment of prostate cancer”.

Reference lists of relevant articles were also manually reviewed to identify additional studies.

Blood-based biomarkers

Blood is the most common and easiest means of determining abnormalities in patients [20]. Different blood-based PCa biomarkers are discussed below (Table 1).

PSA

PSA or kallikrein-3 discussed above is a glycoprotein enzyme encoded by the KLK3 gene in humans [21]. Its function is to promote sperm motility [21]. One of the twentieth century's most significant discoveries was the discovery of PSA as the reduction in PCa mortality observed in the last decade is mostly attributed to this test [22]. PSA was first approved by FDA in 1986 as a test to aid the management of patients diagnosed with PCa [23]. It can exist in free or bound/complexed form [24]. Majority of PSA detected in serum is bound to protein. A total PSA measurement includes both complexed PSA and inactive free PSA. PSA levels can also rise in patients due to prostatitis (inflammation of the prostate), BPH (enlargement of the prostate), or urinary infection [25].

PSA count higher than 4 ng/ml is associated with a PCa specificity of 94% [26]. Currently there is great emphasis on early diagnosis of PCa for maximizing survival rate. Stage 1 PCa which is confined to just one area of prostate corresponds to PSA level 10ng/ml [26]. PSA levels are higher in higher grade cancers. Therefore, the test's ability to accurately screen for PCa is hindered by its low sensitivity [26]. To avoid over diagnosis due to PSA test results, numerous

tests have evolved to find an appropriate, improved blood-based diagnostic test.

4 K score

The 4 K score has had limited success as a diagnostic tool as only 20 research articles have been published from 2014 to 2023 (Fig. 4). It is a blood-based test that is used to detect high-grade PCa (Gleason > 7) in patient having high PSA levels [27]. PSA though an effective test, lacks specificity for high-risk aggressive PCa. The 4 K score test, however, offers specificity and sensitivity for the detection of aggressive PCa. It measures the level of four kallikrein protein biomarkers (total PSA, free PSA, PSA level, and human kallikrein-related peptidase (hk2)) and combines these parameters with the clinical information based on DRE outcome into an algorithm (range 0–100) [28]. This is a non-invasive test, which reduces overtreatment and unnecessary biopsies, and predicts long-term risk of distant metastasis [29, 30].

PHI (PCa index)

Prostate Health Index or PHI was approved in 2012 by the US Food and Drug Administration [31]. A total 109 research articles that have been published are present in PubMed from 2001 to 2023 for development of PHI indicating its usefulness as a PCa biomarker (Fig. 4). PHI is calculated using a mathematical formula that combines total PSA, free PSA, and $[-2]$ proPSA into a single score $PHI = ([-2] \text{ proPSA/free PSA}) \times \sqrt{\text{PSA}}$ [32]. Men with higher $[-2]$ proPSA and total PSA, with a lower value of free PSA are more prone to PCa. The measurement of % $[-2]$ proPSA improves the accuracy of PCa detection in comparison with PSA or % fPSA, particularly in the group of patients with PSA between

Table 1 Summarizes blood and urinary biomarkers

TEST	Sensitivity %	Specificity %	Advantage	Disadvantage	References
Blood-based biomarker					
1. PSA	87.5	60–70	Early detection	False positives are common	[35]
2. 4 K score	94	–	A well-calibrated algorithm for predicting risk of high-grade PCa prior to biopsy	Includes PSA for interpretation	[25]
3. PHI	92	30	High accuracy for detecting PCa and discriminating between aggressive and non-aggressive PCa	Uncertain reference range and ethnic variability, this also includes PSA	[36]
Urinary biomarker					
4. PCA3	73	65	High overexpression in PCa makes it an excellent biomarker	Uncertain cut-off value	[37]
5. Select MDx	76.9	49.6	Provides risk score aggressive prostate cancer	Include clinical variables	[38]
6. ExoDx	92	–	Does not require DRE, uses an algorithm independent of clinical features to provide a risk score that discriminates benign/low-grade PC and high-grade PCa		[39]

2 µg/L and 10 µg/L [33]. According to the research done by White and colleagues, the PHI test can save unnecessary biopsies by up to 40% [32] as it identifies the patients that actually need a biopsy [34].

Urinary biomarkers

The urine-based biomarkers aim to overcome the shortcomings of serum-based biomarkers and to reduce unnecessary biopsies. The major advantage of urinary biomarkers is that they are non-invasive compared to the more conventional blood test or biopsy [40]. These urine-based biomarkers are useful in early detections as well as in identifying aggressive tumors for radical treatment. Some of the urine-based biomarkers are described below.

PCA3

PCa Antigen 3 (also known as DD3) was the first urine-based biomarker discovered in 1999 by Marion Bussemakers [41, 42]. PCA3 has been explored widely as a biomarkers indicated by 233 research articles published between 2005 and 2023 (Fig. 3). Its overexpression in > 90% PCa cases makes it an excellent biomarker [43]. This test detects PCA3 mRNA and PSA mRNA in the first urine samples collected after prostate massage following DRE. A PCA3 test result below 20 indicates a minimal likelihood of significant cancer, with less than 15% probability. In this case, it may be suitable to consider re-examination after a period of 6–12 months. This approach can enhance patient comfort and minimize the potential complications associated with prostate biopsy procedures [44]. On the other hand, a PCA3 test value exceeding 35 suggests an elevated risk of PCa and warrants further evaluation through a prostate biopsy procedure [45].

Select MDx

Select MDx test is a urine-based test developed by The Netherlands: Nijmegen Laboratory in 2016. It is probably not used very widely as only 35 research articles were published in PubMed from 2017 to 2023 [38]. Select MDx is a reverse transcription PCR (RT-PCR) assay, performed using urine sample collected immediately following DRE that measures the urinary mRNA levels of DLX1 and HOXC6 biomarkers [46]. This test is useful in determining whether a patient needs to undergo prostate biopsy in future or not. Elevated levels of DLX1 and HOXC6 mRNA in patient urine samples indicate increased probability for high-grade PCa Gleason score > 7 (GS > 7) [47, 48]. This test has an accuracy of 98% for detecting significant PCa (Gleason 3 + 4 and higher) [49].

TMPRSS2-ERG fusion

The TMPRSS2-ERG fusion test detects the gene re-arrangements between the TMPRSS2 (transcriptional promoter transmembrane protease serine 2) and ERG (the oncogene erythroblast transformation-specific-related gene) both located on chromosome 21 in PCa patients. The prevalence of TMPRSS2-ERG was 30–50% in patients with localized prostate cancer. TMPRSS2-ERG was also detected in urine after DRE from men with PCa [50]. Eight of nineteen patients with PCa (43%) had detectable TMPRSS2-ERG transcripts in their urine [50]. This high level of ERG gene is directly linked with the shorter life span, aggressive stage of cancer, metastasis, and a high Gleason scores [51].

ExoDx prostate IntelliScore (EPI)

EPI is a diagnostic tool that utilizes expression levels of three genes in exosomes to accurately determine likelihood of high-grade PCa starting from Grade 2 or higher [52]. EPI is an independent test which does not include PSA or any derivative of it in its testing algorithm [39]. It measures exosomal RNA markers in urine sample, allowing clinicians to use test along with other clinical variables such as clinical nomograms, mpMRI, or standard-care risk calculators. In fact, use of EPI urine exosome test has shown to reduce unnecessary biopsies by 27% compared to standard care methods [52].

Tissue-based biomarkers

The confirmatory test for PCa is a tissue biopsy. Once a biopsy is done, histopathology is performed. The histopathological assessment of stained tissue sections is ultimately used for critical diagnosing, characterizing, and managing PCa [53]. Histological grading of adenocarcinoma of prostate, based on Gleason grades provided by International Society of Urological Pathology, is the most used prognostic indicators for PCa [53].

Although histopathology offers reasonable degree of predictive ability, advances in molecular biology have given birth to tissue-based biomarkers that provide physician additional information about a given patient's risk and treatment options [54]. Of the multitude of tissue-based biomarker assays that have been developed, the most valuable are Oncotype DX, Prolaris, and ProMark [55] which are discussed below.

Prolaris

It is a tissue-based test for patients having a low or intermediate risk of PCa offered by Myriad Genetics Inc [6]. This

test is based on the expression pattern of 13 genes involved in cell cycle progression (CCP) and 15 housekeeping genes [56]. CCP score is expressed between 0 and 10 [54]. An increase in each unit indicates the doubling of gene expression level suggesting a more aggressive tumor. The CCP value is checked in males recently diagnosed with PCa (Prolaris biopsy test) or a man who has already undergone prostatectomy (Prolaris prostatectomy test) [54]. A high CCP value suggests a highly aggressive tumor and a lower CCP value indicates tumor that is less aggressive [38]. The CCP value is a robust prognostic indicator, providing valuable information, distinct from conventional clinicopathologic factors like PSA and Gleason score.

ProMark

It is a quantitative immunofluorescence assay used to predict aggressiveness of PCa in tissues [51]. The major role of ProMark is to differentiate between patients for active surveillance (AS) and those who require radical prostatectomy (RP). Using biopsy samples, this test can accurately stratify high- and low-risk cancer. It is a protein- based assay that measures level of eight proteins: CUL2, SMAD4, PDSS2, HSPA9, FUS, pS6, YBOX1, and DERL1 [54]. These eight proteins individually correlate with tumor aggressiveness and ProMark quantitatively analyzes expression level of these protein which predict aggressiveness of the disease. A risk score is calculated on the scale of 0–1 depending on the levels of these eight proteins. A risk score of <0.03 is “favorable” and risk score of >0.08 is “unfavorable” [54, 57].

The current prognostic markers discussed above rely on risk stratification systems that take into account clinicopathological variables such as Gleason score, pathological tumor stage, and serum PSA levels at the time of diagnosis [58]. These indicators are effective in assessing disease severity and correlating with patient survival rates [59]. Besides the risk stratification systems, there are several multi-omic biomarker panels currently under investigation for prognostication purposes. These include the OncotypeDX Genomic Prostate Score, Decipher, and Prolaris tests [60]. Research studies demonstrate that these panel-based tests often achieve an area under the ROC curve (AUC) higher than 0.7. However, it is worth noting that these tests are not widely accessible at present [61]. They are only available commercially through select private clinics which greatly limits their affordability and availability outside of certain geographic locations [19].

New generation biomarkers

In the pursuit of precision medicine for patients, researchers are pioneering new diagnostic techniques. These include studying alterations in novel biomarkers like mRNAs,

microRNAs, lncRNAs, and SNPs. Development of these biomarkers will help in diagnosing the aggressiveness of PCa, thereby enhancing their prognosis and quality of life. In the following section, we will discuss some molecular biomarkers that have recently been reported.

MicroRNA

MicroRNAs (miRNAs) are being studied extensively as potential alternative molecular markers for PCa [18]. These are a group of small, non-coding RNAs that play a significant role in gene regulation [62]. They appear to play pivotal roles in various cellular processes, including proliferation, differentiation, migration, apoptosis, stem cell maintenance, and metabolism [63]. Notably, miRNAs can be detected in various body fluids with tissue-specific expression patterns and are dysregulated in different type of cancer. In addition, miRNA can be used to classify tumors based on their lineage of origin, stage of differentiation, and aggressiveness. As a result, these miRNAs offer great potential as biomarkers for PCa diagnosis [18] and for developing precision medicine therapy as they regulate different signaling pathways.

The miRNA expression pattern in PCa patients differs from that of the control group, making miRNA a valuable diagnostic marker. Studies conducted by Mitchell et al. first identified serum miR-141 as a diagnostic marker for PCa. In their study, they examined six different miRNAs and found that miR-141 displayed the highest level of differential expression with a 46-fold overexpression in serum of PCa patient samples. This particular miRNA exhibited excellent accuracy in distinguishing advanced metastatic and normal PCa patient samples, achieving a sensitivity rate of 100% and specificity rate of 60% [64]. In a study involving over 300 patients, Li et al. reported downregulation of miRNA95 in non-metastatic tumors and its correlation with the postoperative T/N state. They suggested that miRNA-23b, miRNA-95, miRNA-143, and miRNA-183 could serve as potential biomarkers for aiding in the diagnosis and prognosis of prostate cancer [66]. Diagnostic tests related to miRNA, such as the miRview™mets test, were initially used to pinpoint the specific origin of tumor cells by analyzing miRNAs. The first version evaluated a set of 48 miRNAs in tissue using qPCR and could distinguish between 25 different types of tumors. Subsequently, the second generation, known as miR-view mets2, expanded the panel to 64 miRNAs and identified 49 types of tumors [65]. To date, there has not been a diagnostic or prognostic test developed solely based on detecting miRNA in bodily fluids [65]. A few diagnostic miRNA markers are listed in Fig. 5.

miRNAs can detect cancer at early stage. For example, Guo et al. conducted a study on miR-155 gene expression in tumor tissue involving 86 PCa patients and found that the level of miR-155 was significantly elevated in tumor

tissue compared to para-carcinoma tissue ($p < 0.05$). In addition, they observed a positive correlation between PSA and miR-155 expressions with TNM stage and tumor volume ($p < 0.05$), suggesting that the combination of PSA and miR-155 could aid in the early detection of PCa [67]. Wani et al. reported miR-2909 in urine samples as a potential diagnostic marker, as its levels were found to be elevated in PCa patients ($n = 90$) compared to control groups, BPH ($n = 10$), and healthy individuals ($n = 50$). Further, the expression profiles of miR-21, miR-574, and miR-141 have been utilized for diagnosing early-stage prostate carcinoma from urine exosomes using a lectin-based agglutination method [68]. Studies by Martin et al. in 2018, profiled 372 miRNAs in plasma from PCa patients and healthy controls, before and during the commencement of treatment. Four miRNAs (miR-4289, miR-326, miR-152-3p, miR-98-5p) exhibited elevated levels in the PCa patients' plasma. Study showed that overexpression of miR-152-3p led to increased cell proliferation and migration, suggesting its direct involvement in cancer progression [69]. The study presents a biomarker panel of these four miRNAs, particularly highlighting miR-152-3p's link to cancer-related pathways, as a promising tool for the improved early diagnosis of PCa [69].

Other miRNAs can detect advance-stage PCa. Studies have revealed miR-32 to be consistently dysregulated in PCa, while overexpression is observed at the CRPC stage [70]. PCa cells show increased proliferation and decreased apoptosis due to overexpression of miR-32 [71]. miR-32 is an AR-regulated miRNA and it contributes to PCa cell growth and progression by blocking the expression of the tumor-suppressor genes phosphoinositide-3-kinase interacting protein 1 (PIK3IP1) and B-cell translocation gene 2 (BTG2), favoring the PI3K/AKT/mTOR pathway [71] (Table 2). A study by Barcelo et al. in 2019 highlighted the potential of investigating semen exosomes miRNAs for PCa prognosis

and diagnosis. They found that combining standard PSA tests with miR-142-3p, miR-142-5p, and miR-223-3p could differentiate between PCa patients and those with benign prostatic hyperplasia (BPH) with an AUC of 0.82. In addition, combining PSA with miR-324-3p and miR-374b-5p could distinguish Gleason scores of 7 or higher [72].

miRNAs can also be used as prognostic biomarkers. Cuzick et al. conducted a study using qRT-PCR to analyze the prognostic value of miRNA expression related to cell cycle progression, showing significant potential for predicting prognosis and guiding treatment decisions. They identified miR-1290 and miR-375 as predictive biomarkers in Castration-resistant PCa patients [73, 74]. Furthermore, Penney et al. found that assessing the expression of a 157-gene signature could enhance our understanding of tumor de-differentiation process, aiding in predicting Gleason score and relative risk for guiding therapy decisions more effectively while reducing overtreatment. Both miR 182-5p and miR-375-3p in plasma samples from patients were also discovered to be valuable as prognostic and screening biomarkers for PCa [75].

Long non-coding RNAs

Long non-coding RNA (lncRNAs) are a type of RNA that does not have the ability to code for proteins [76]. Several studies have suggested that lncRNAs can play a role in cell proliferation, tumorigenesis, and malignant transformation [77] and hence their potential role they as tumor-suppressor genes and oncogenes is being studied [78]. Recently, their potential role as molecular biomarkers has been explored [77]. lncRNAs including PCA3, SChLAP1, and PCAT1 have been suggested as promising biomarker candidates for PCa due to their distinctive expression patterns in PCa patients [77] (Table 2).

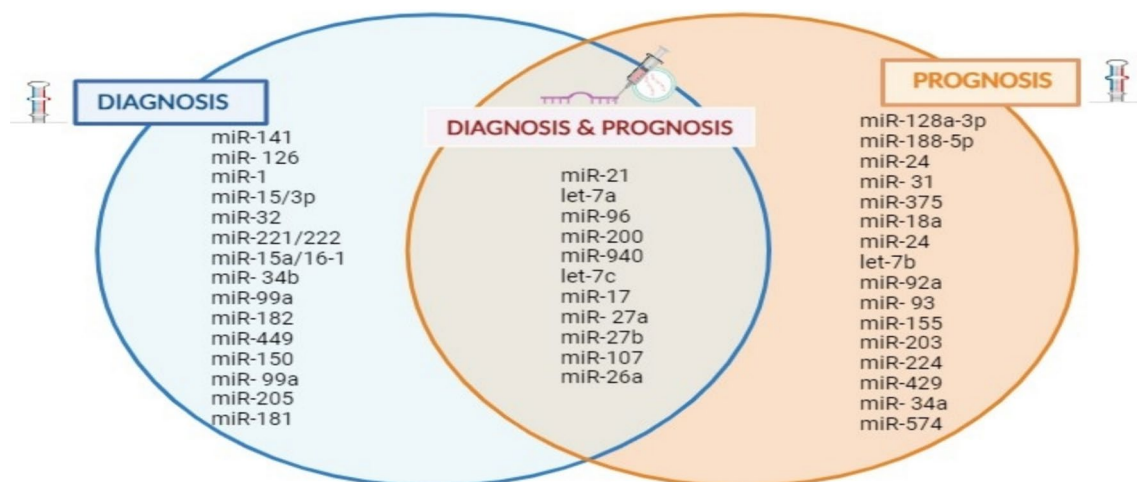


Fig. 5 Different diagnostic and prognostic miRNA biomarkers in PCa

Table 2 Different miRNA and lncRNA biomarkers to distinguish between localized, metastatic, and advanced PCa

	Function	Specimen	Used in	Distinguishes between	References
miRNAs					
miR-21	Enhances tumor invasiveness and triggers the development of castration-resistant phenotype	Serum, urine	Diagnosis, prognosis, therapy	Localized/local versus advanced PCa cases	[66]
miR-221/miR-222	Promote cellular growth, invasion, survival, increase the ability to form colonies and enhance tumor formation	Serum	Diagnosis, prognosis	Metastatic and localized/local advanced PCa	[97]
miR-141	Is important in androgen-dependent and in metastatic castration-resistant	Blood	Diagnosis	Metastatic and normal PCa patients	[64]
miR-375	Is important for an early diagnosis	Serum	Diagnosis, prognosis	Metastatic CRPC and localized PCa patients	[64]
LncRNAs					
PCA3	Overexpression of PCA3 contributes to modulation of AR signaling in tumor cells	Urine	Diagnosis, prognosis	Metastatic and normal PCa patients	[45]
SChLAP1	Antagonizes the SWI/SNF-complex	Tissue	Diagnosis, prognosis	Metastatic and benign PCa patients	[72]
PCAT1	Promotes cell proliferation, apoptosis, migration, invasion, epithelial–mesenchymal transition and cancer progression	Tissue	Prognosis	Normal and localized/metastatic PCa	[73]

PCA3, as already discussed above in urinary biomarker section, is a specific PCa biomarker. Studies by Lemos et al. demonstrate that PCA3 knockdown in LNCaP cells inhibits AR signaling, cell growth, and viability. This suggests that overexpression of PCA3 may contribute to the modulation of AR signaling in tumor cells [79]. Studies by Sanda et al. [82] demonstrate that combination of PCA3 and TMPRSS2:ERG fusion results in improved specificity for detecting PCa with GS ≥ 7 [80]. The main advantage of PCA3 is that it reduces unnecessary prostate biopsies.

SChLAP1 (second chromosome locus-associated with prostate-1) is a lncRNA that exhibits high expression in 25% of primary PCa cases [74]. Its expression is higher in CRPC PCa, and its presence has been associated significantly with an increased risk of biochemical recurrence, clinical progression, metastasis, and PCa mortality [74, 81]. SChLAP1 interacts with the SWI/SNF (switch-sucrose non-fermentable) complex to facilitate chromatin remodeling, effectively counteracting the tumor-suppressor effects mediated by SWI/SNF [74]. Expression analysis using RNA in situ hybridization by Mehra et al. indicates that levels of SChLAP1 can independently predict biochemical recurrence following radical prostatectomy [83]. Moreover, Mehra et al. suggests a correlation between SChLAP1 expression and PCa lethal progression

[83]. Hence, SChLAP1 holds great potential as a valuable clinical biomarker for predicting metastasis and biochemical recurrence in PCa patient.

PCAT1 (prostate cancer-associated ncRNA transcript 3) is an overexpressed lncRNA that was initially identified in PCa through RNA-sequencing analysis [81]. It is predominantly expressed in the prostate and is found to be upregulated in both high-grade localized and metastatic forms of PCa. It is associated with various cellular processes such as cell proliferation, apoptosis, migration, invasion, epithelial–mesenchymal transition, and cancer progression via the Wnt/ β -catenin signaling pathway [75]. The expression of PCAT-1 is intricately linked with extent of infiltration, lymph node metastasis, and TNM staging. In 2011 Prensner et al. reported that PCAT1 negatively regulates BRCA2 tumor-suppressor protein and positively regulates Myc oncoprotein [81]. As a result of these findings, PCAT1 is considered a potential biomarker for prognosis and prediction of prostate cancer outcomes.

These observations support the notion that lncRNAs have promising applications as molecular biomarkers in managing prostate cancer. Also role of lncRNA in regulating different pathways makes it a suitable candidate for precision medicine.

Single nucleotide polymorphism (SNPs)

Single nucleotide polymorphism or SNPs are most common type of genetic variation among people. These variations occur in the DNA sequence, in approximately 1 out of 800 base pair [84]. The major role of SNPs in PCa is that it helps in risk prediction and enhances fine mapping based on pathophysiology of disease [85]. There are various genetic polymorphisms that have been proposed as potential prognostic biomarkers for prostate cancer in clinical practice. One such polymorphism, rs1400633 in the MSH2 gene, has shown a significant association with more aggressive disease characteristics [86]. This gene encodes a core protein involved in DNA mismatch repair pathways responsible for detecting and correcting base pair mismatches during replication [86]. The rs4648302 variant has been identified as a potential biomarker for disease recurrence in prostate cancer [87]. This single nucleotide polymorphism is located in 3'-untranslated region of PTGS2 gene, which encodes enzyme prostaglandin-endoperoxide synthase 2. The presence of this polymorphism could serve as an independent risk indicator for recurrent PCa [87]. Studies by Manholini et al. report that variants of ATM (rs587781894), TP53 (rs786203436 and rs587780075), SPOP, and FOXA1 genes correlate with a poor prognosis [88]. These gene mutations are associated with PCa development and progression [89]. Therefore, SNPs located within these genes show promise as prognostic biomarkers for improving patient management in PCa cases [90]. Another example is polymorphism in 8q24 gene which is linked with increased PCa susceptibility and could be used as a biomarker in PCa diagnosis [85]. The characterization of different SNPs and their roles in PCa and disease progression makes them promising potential candidates as biomarkers that can improve early diagnosis and treatment.

Imaging biomarkers

Pathologic staging and grading identified through imaging techniques can also serve as biomarkers, influencing the prognosis of PCa. Magnetic resonance imaging (MRI) is widely used to diagnose, assess risk, locate the cancer, and determine the stage in PCa. Iglesias et al. investigated the potential use of imaging biomarkers such as radiomic, diffusion, and perfusion features derived from MRIs. Gray-Level Co-Occurrence Matrix (GLCM) is a standard statistical method for analyzing texture of tumors. Radiomic features such as GLCM_Inverse Difference Moment, GLRLM_Run Variance, GLCM_Inverse Difference, Skewness and Neighboring grey tone difference strength were higher in patients experiencing recurrence 10 years after diagnosis compared to those who did not [93]. Decrease in certain

radiomic features such as Major Axis Length, Maximum 2D Diameter, and Maximum 3D Diameter were observed in patients experiencing relapse in different prostate regions including seminal vesicles, as well as in the whole prostate [93]. Recent studies have shown that PSA screening with additional post-screening MRI can effectively reduce overdiagnosis [94]. In 2021 Eklund et al. demonstrated that performing MRI on men with abnormal PSA results led to a significant decrease in the requirement for biopsies and related risks [95]. Nordström et al. discovered that using the Stockholm3 biomarker test alongside an MRI-targeted biopsy approach reduced over-detection without compromising the ability to identify clinically significant cancers [96]. However, it is crucial to note that low-grade and smaller tumors are often not detected by MRI [94].

Conclusion and future directions

PCa diagnosis and therapy has come a long way since the discovery of phosphatic acid phosphatase by Huggins and Hodges in 1941. Though the number of patients diagnosed went up slightly between 2015 and 2019 (dipped again in 2020), yet percentage of death due to disease has reduced slowly but steadily (National Cancer Institute). This has been possible only because of wide-scale screening of men above a certain age group, using biomarkers, especially PSA. No doubt PSA has revolutionized PCa detection.

However, many problems persist in diagnosis and treatment of PCa which should likely be resolved by development of appropriate biomarkers. Biomarkers for early diagnosis are still few and suffer from lack of sensitivity or specificity. A biomarker which correlates with uncontrolled prostate cell proliferation, is secreted in blood, and can be detected before PSA values increase above 4 ng/ml would be ideal. The non-invasiveness and cost of these markers would also be important in their widespread adoption as diagnostic tools. GPCRs [92] and other genes that are overexpressed in PCa such as Dlx1 [91] should be explored further for their use in PCa diagnostics.

Main utility of tissue biomarkers is depicted in analyzing cancer aggressiveness and identification of perturbed pathway leading to cancer. Mutations in several pathways can ultimately lead to PCa. For precise treatment, it is important to identify perturbed pathway in the patient. The commonly altered pathways include AR signaling pathway (70%), PI3K/AKT/mTOR signaling (40–60%), NF- κ B signaling pathway (16.3%), JAK/STAT signaling pathway (10–30%), HER2/neu pathway (60%), GPCR-mediated pathway (30%), and Wnt signaling pathway (22%) (Fig. 6). Identification of mutation and pathways have direct implications on treatment mechanisms adopted and patient survival. Using a combination of genomics, transcriptomics, and AI approaches, it

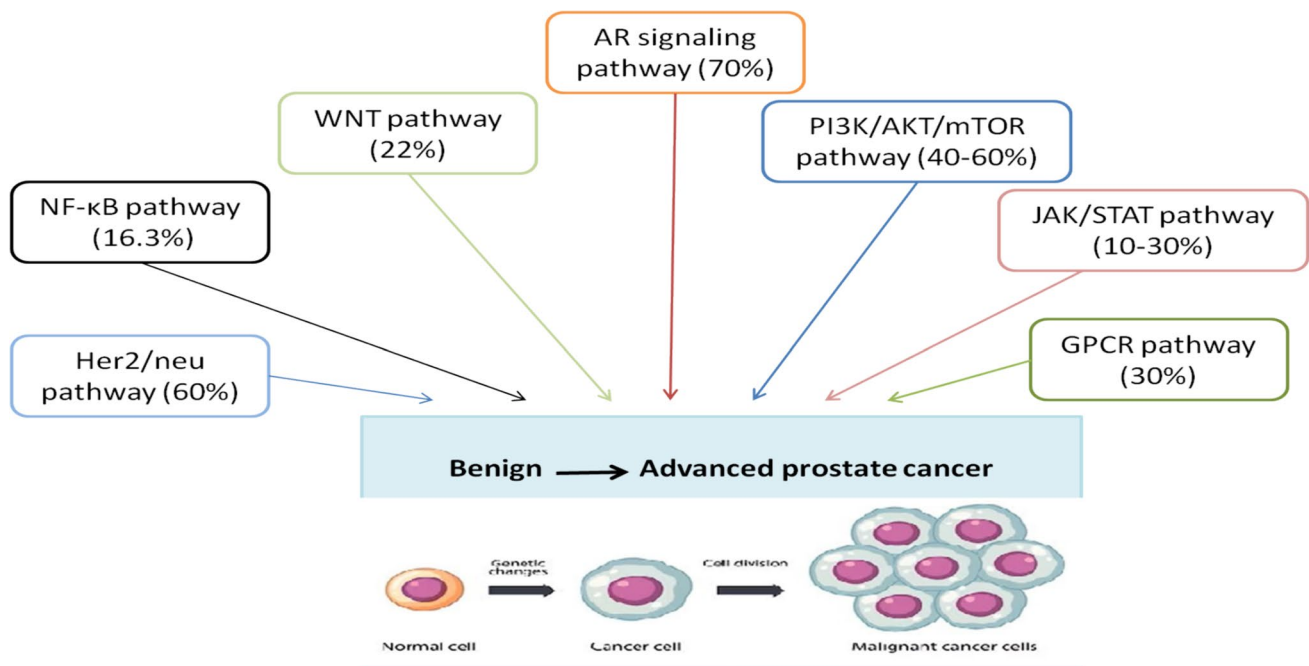


Fig. 6 Pathways leading to prostate cancer development and progression

would be useful to develop novel algorithms that integrate information regarding gene mutations related to specific pathways and clinical progression and aggressiveness of disease. This would enable clinicians and patients to make informed choice regarding treatment options and disease outcome.

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Declarations

Conflict of interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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